RouteMind AI

Tell us the shipment.
We’ll find the lane.

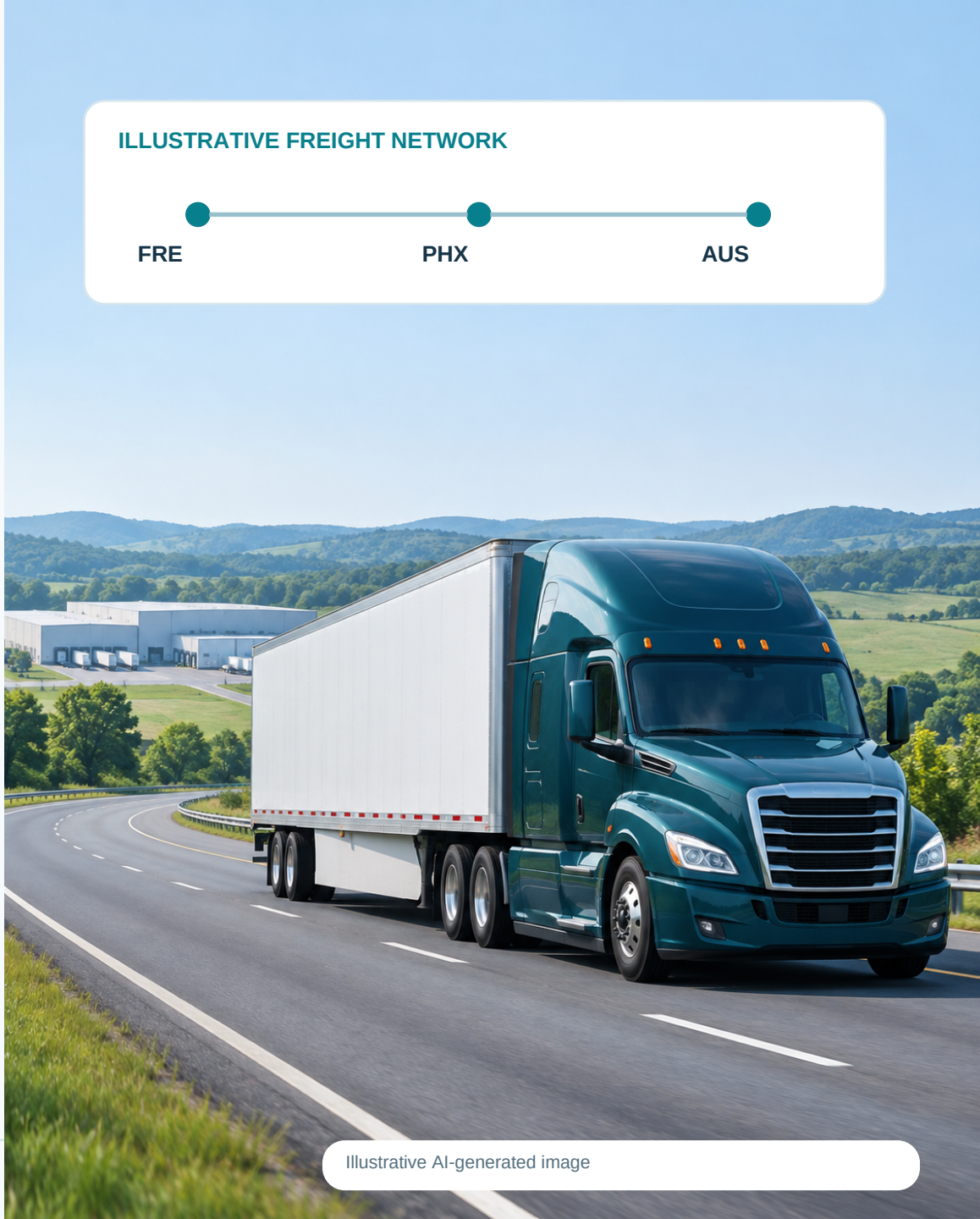
Conversational ground-freight planning with evidence and verified route decisions.

PROJECT TEAM

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What we will show

The problem, the data, the approach and how we check the results.



The project

Turn a shipment request into feasible, ranked routes.



The data

A reproducible network and versioned document corpus.



The approach

NLP + hybrid RAG + deterministic optimization.



Evaluation

Check routing, language extraction, retrieval and answer grounding.

Understand the request. Verify the route. Make the supporting evidence visible.

Which route would you choose?

15 pallets of fragile electronics. Fremont to Austin. Budget: $4,000.

OPTION A / LOWER COST



$2,100

One transfer

OPTION B / EARLIER ARRIVAL



$3,400

Direct · Arrives 8 hours earlier

Before choosing, what else would you need to know?

Meet RouteMind AI

A conversational assistant for comparing suitable ground-freight routes.

Describe

Locations, quantity, budget, deadline and cargo care.

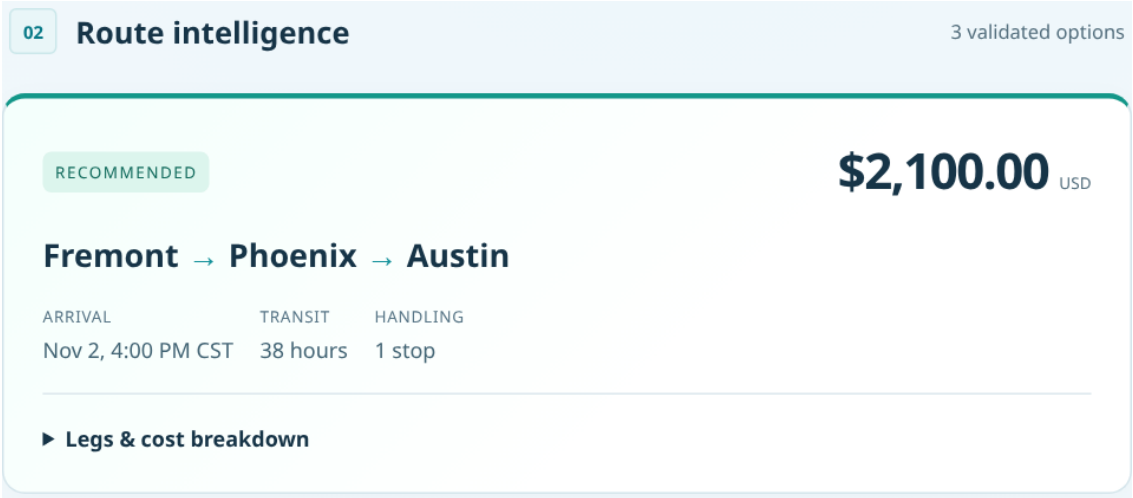
Compare

Review cost, arrival, transfers and supporting information.

Explore

“Make it faster, but keep the budget.”

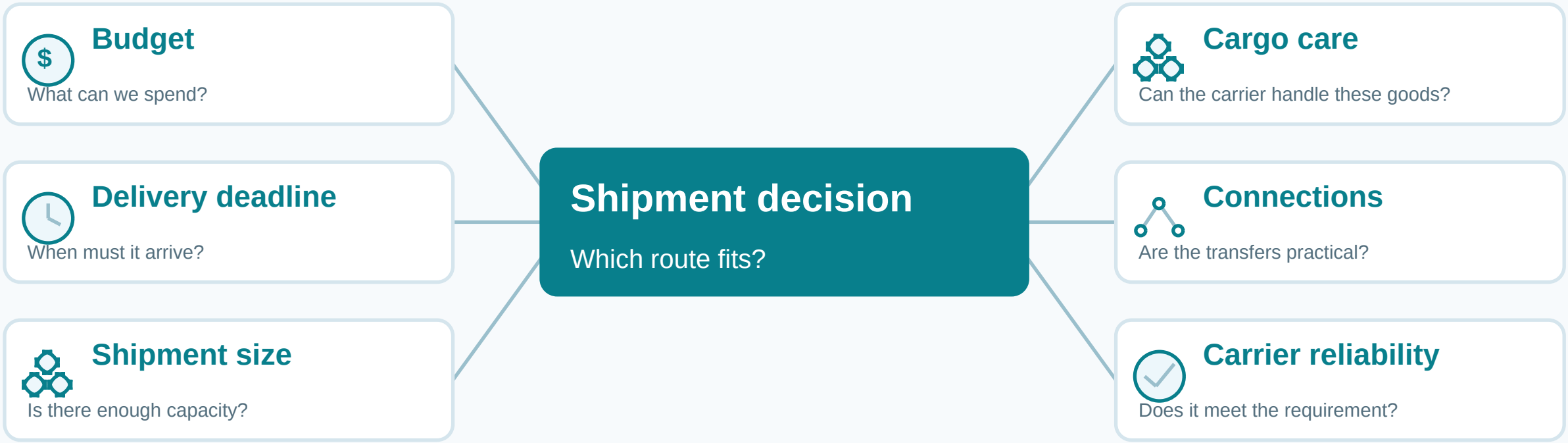
ACTUAL APPLICATION / SYNTHETIC SHIPMENT



The user reviews the options and makes the final choice.

What makes a route suitable?

A route must satisfy the shipment’s requirements—not just offer a low price.



A change to one requirement can change the best route.

Fragile electronics: Fremont to Austin

15 pallets • $4,000 ceiling • Fixed demo start: November 1, 2026, 06:00 UTC.

Priority	Route	Total cost	Arrival from demo start
Cheapest	Fremont > Phoenix > Austin	$2,100	+40 hours
Fastest	Fremont > Austin (direct)	$3,400	+32 hours

Follow-up: “Make it faster, but keep the budget.”

The revised request preserves unchanged requirements and compares the verified outcomes.

Synthetic, reproducible and explicitly labelled

Operational facts are authored or generated in app/seed.py; no live carrier feed is connected.

Dataset	Size	Contents / provenance
Network hubs	18	US hub names, timezones and authored transfer rules
Carriers / capability flags	10 / 40	Fictional profiles; four explicit capability flags per carrier
Scheduled services	180	Fixed-seed departures, arrivals, capacities and prices
Evidence corpus	58	50 carrier passages + 4 handling notes + 4 planning cases
Quality benchmark	64 NLP / 50 policy	Synthetic labels; separate development/test groups

SQL: operational records | ChromaDB: document vectors | No private customer data or live quotes

Language enters. Verified decisions leave.

LangGraph coordinates the planning workflow; FastAPI serves the web application.

01

Interpret

Rules or typed Gemini/OpenAI

02

Clarify

Validate fields; ask for gaps

03

Retrieve

MiniLM + BM25 carrier evidence

04

Optimize

Bounded scheduled path enumeration

05

Verify

Recompute from authoritative SQL

06

Explain

Ranked plans + cited context

Two stores, two responsibilities

PostgreSQL/SQLite determines eligibility. ChromaDB supplies supporting text; it cannot approve a route.

Separate interpretation from mathematical correctness

Natural-language flexibility is useful; route feasibility must remain reproducible.

INTERPRETATION

NLP and conversation

Extract fields; ask about ambiguity. Apply contextual edits while retaining other constraints.

OPTIMIZATION

Exact bounded search

Enumerate acyclic paths up to 3 legs. Check budget, timing, capacity, capabilities, reliability and stops.

Independent verification recomputes every returned plan from SQL schedules, prices and transfer rules.

Search by meaning and words, then check the evidence

Policy Q&A selects exact source passages; it does not generate authoritative carrier rules.

HYBRID SEARCH

Retrieve

MiniLM semantic search + BM25 keyword search; merge with RRF.

EVIDENCE SAFEGUARDS

Review

Check relevance, effective dates, revisions and reviewed capability conflicts.

GROUNDED OUTPUT

Cite or abstain

Validate source IDs and quotations. Report insufficient or conflicting evidence.

RRF(d) = sum[1 / (60 + rank(d))]

A correct quotation is not proof that it completely answers the question.

Does hybrid retrieval handle paraphrased questions better?

Test how changing the wording of the same carrier-policy question affects evidence retrieval.

MOTIVATION

Same intent, new words

Users may ask the same policy question with exact terms or everyday language.

METHODS

Three search strategies

Compare BM25 keywords, MiniLM semantics, and hybrid RRF on the same corpus.

HYPOTHESIS

More robust rankings

Hybrid may lose less accuracy on paraphrases while keeping exact-term matches.

Same policy fact. Two ways of asking.

Measure ranked evidence accuracy—not generated answers or route decisions.

A controlled, paired-question experiment

Keep the source corpus, carrier scope, and retrieval depth fixed for all three methods.

01

Select facts

30 labelled policy facts

02

Write pairs

Exact terms + paraphrases

03

Retrieve

BM25 / MiniLM / hybrid RRF

04

Rank docs

Same corpus, same Top-K

05

Score

Recall@1/5, MRR, nDCG@5

06

Analyze

Score drops + error cases

Paired evaluation, fair comparison

Keep both wordings in the same dev/test group; report metric changes and retrieval errors.

Ragas evaluation of carrier-policy answers

An LLM judge evaluates saved policy reports using four complementary metrics.

Metric	Evaluation question	Evaluation inputs
Context precision	Are useful passages ranked first?	Question, context, reference
Context recall	Does the evidence cover the expected facts?	Question, context, reference
Answer relevancy	Does the answer address the question?	Question and answer
Faithfulness	Are answer claims supported by the evidence?	Question, context, answer

Method: score saved reports with reviewed references for the context metrics.

Report skipped and failed cases. Exact quotes can be faithful but irrelevant.

Useful regression evidence, with clear limits

Latest verified software run: 107 tests passed on each of SQLite and PostgreSQL.

Quality-suite result	Test: Hash	Test: MiniLM
Offline rules entity F1 / clarification accuracy	1.00 / 1.00	1.00 / 1.00
Answerable policy retrieval recall	0.90	0.90
Relevance-gate accuracy proxy	0.96	0.96
Live-model / human grounding quality	Not measured	Not measured

Shared templates and near-duplicate passages make these tests optimistic.

MiniLM development gate proxy: 0.88 — semantic similarity can still admit irrelevant text.

An auditable planning assistant, with a defined boundary

Working prototype today; independent quality evidence and real integrations are future work.

DELIVERED

Demonstrated

Conversational planning, cited policy passages, comparisons and a read-only explorer.

EVALUATION PRIORITY

Next to validate

Independent labels, live-model evaluation, human grounding and gate calibration.

EXPLICIT LIMITS

Outside today's scope

Live rates, traffic, bookings, real capacity and production multi-user operation.

Demo: describe a shipment > revise its priority > inspect a policy source.

Repository: github.com/chetsp/RouteMind-AI